INCH-POUND
MIL-DTL-24643/24D
27 November 2023
SUPERSEDING
MIL-DTL-24643/24C
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +302 DEGREES FAHRENHEIT (+150 DEGREES CELSIUS), 300 VOLTS, TYPES LSTCJX, LSTCKX, LSTCTX, LSTCJXN, LSTCKXN, AND LSTCTXN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in the following variations: LSTCJX, LSTCKX, and LSTCTX (watertight with circuit integrity), and LSTCJXN, LSTCKXN, and LSTCTXN (non-watertight with circuit integrity).

Construction
Types LSTCJX, LSTCKX, and LSTCTX (Watertight with Circuit Integrity)
Types LSTCJXN, LSTCKXN, and LSTCTXN (Non-Watertight with Circuit Integrity)

First	-	Stranded conductor, uncoated (see table I or table II for size and stranding).
Second	-	Extruded silicone rubber insulation (see table I or table II for dimensions).
Third	-	Glass braid (see table I or table II for dimensions).
Fourth	-	Transparent braid covering.
Fifth	-	Two conductors cabled together to form a pair with a lay not greater than 2.25 inches. For LSTCJX and LSTCJXN, one iron conductor and one constantan conductor. For LSTCKX and LSTCKXN, one chromel and one alumel conductor. For LSTCTX and LSTCTXN, one copper conductor and one constantan conductor. Types J, K, and T thermocouple conductors shall be in accordance with ASTM E230/230M. Color code applied by method 4; see table III for color code sequence.
Sixth	-	The specified number of pairs (see table I or table II) shall be cabled together with a lay not greater than 10 inches. Fillers may be used as necessary to form a firm, well-rounded assembly. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward.
Seventh	-	A binder applied helically with overlap.
Eighth	-	Silicone rubber jacket colored orange or red (see table I or table II for thickness). Jacket surface marking is not applicable.
Ninth	-	Braided metal armor.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

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TABLE I. Details (watertight).

Military part no. M24643/24	Type and size	Conductors				Overall cable (inch)	
		No. of pairs	No. of strands and strand diameter (inch)	Diameter over insulation nom. (inch)	Diameter over braid covering max. (inch)	Jacket thickness (min. avg.)	Diameter (max.)
-01AN	LSTCJX-3	3	7/0.0201	0.100	0.125	0.050	0.742
-02AN	LSTCJX-7	7	7/0.0201	0.100	0.125	0.062	0.983
-03AN	LSTCJX-12	12	7/0.0201	0.100	0.125	0.062	1.269
-04AN	LSTCKX-1	1	7/0.0201	0.100	0.125	0.038	0.456
-05AN	LSTCKX-3	3	7/0.0201	0.100	0.125	0.050	0.742
-06AN	LSTCKX-7	7	7/0.0201	0.100	0.125	0.062	0.983
-07AN	LSTCKX-12	12	7/0.0201	0.100	0.125	0.062	1.269
-08AN	LSTCTX-1	1	7/0.0113	0.065	0.085	0.038	0.350
-09AN	LSTCTX-3	3	7/0.0113	0.065	0.085	0.038	0.552
-10AN	LSTCTX-7	7	7/0.0113	0.065	0.085	0.050	0.731
-11AN	LSTCTX-12	12	7/0.0113	0.065	0.085	0.063	0.964

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TABLE II. Details (non-watertight).

Military part no. M24643/24	Type and size	No. of pairs	Conductors			Overall cable (inch)	
			No. of strands and strand diameter (inch)	Diameter over insulation nom. (inch)	Diameter over braid covering max. (inch)	Jacket thickness (min. avg.)	Diameter (max.)
-12AN	LSTCJXN-3	3	7/0.0201	0.100	0.125	0.050	0.742
-13AN	LSTCJXN-7	7	7/0.0201	0.100	0.125	0.062	0.983
-14AN	LSTCJXN-12	12	7/0.0201	0.100	0.125	0.062	1.269
-15AN	LSTCKXN-1	1	7/0.0201	0.100	0.125	0.038	0.456
-16AN	LSTCKXN-3	3	7/0.0201	0.100	0.125	0.050	0.742
-17AN	LSTCKXN-7	7	7/0.0201	0.100	0.125	0.062	0.983
-18AN	LSTCKXN-12	12	7/0.0201	0.100	0.125	0.062	1.269
-19AN	LSTCTXN-1	1	7/0.0113	0.065	0.085	0.038	0.350
-20AN	LSTCTXN-3	3	7/0.0113	0.065	0.085	0.038	0.552
-21AN	LSTCTXN-7	7	7/0.0113	0.065	0.085	0.050	0.731
-22AN	LSTCTXN-12	12	7/0.0113	0.065	0.085	0.063	0.964

TABLE III. Color identification code.

Pair number	Constantan or alumel color	Iron, copper, or chromel leg	
		Base color	Tracer color
1	Red	White	N/A
2	Red	Black	N/A
3	Red	Green	N/A
4	Red	Orange	N/A
5	Red	Blue	N/A
6	Red	White	Black
7	Red	White	Green
8	Red	White	Orange
9	Red	White	Blue
10	Red	White	Red
11	Red	Black	White
12	Red	Green	White

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor continuity	No failure
ASTM E230/230M calibration limits of error:	
Types LSTCJX and LSTCJXN	±3.96 °F (±2.2 °C)
Types LSTCKX and LSTCKXN (-32 to 530 °F)	±7.2 °F (±4 °C)
Types LSTCTX and LSTCTXN	±1.8 °F (±1 °C)

Group A:

Visual and dimensional	No failure
Watertightness types LSTCJX, LSTCKX, and LSTCTX only (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance (insulation, scrapes min.)	250
Crack resistance	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (watertight types only)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	800

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Elongation (percent, min.)	250
Durometer (jacket) – type A (hardness, min.)	60

Group C:

Physicals (cable aging) (500±18 °F [260±10 °C])	
Jacket (cable)	
Tensile strength (lb/in^2, min.)	600
Elongation (percent, min.)	150
Cable fill (sealant) removability (watertight types only)	No failure
Armor (conformance to material, construction, and coverage)	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Gas flame (circuit integrity) (3 hours)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Armor	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Smoke index (max.)	
Jacket	35
Fillers	45
Insulation	35
Toxicity index (max.)	
Jacket	5

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Fillers	5
Insulation	1.5
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (min.)
All types and sizes	100

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-040)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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